Project: Telco Customer Churn - Exploratory Data Analysis and Data Preprocessing

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Course: ITAI 1371

Contribution Journal

Brian- Main Notebook Development and Data Cleaning

I was able to complete most of the initial implementation of the Jupyter notebook and worked on the data cleaning and preprocessing steps. I loaded the dataset, created a backup copy of the original data, displayed sample records, checked the dataset dimensions, and examined the missing values.

I also removed the customerID column because it identifies customers rather than describing a general customer characteristic for predicting churn. I was also able to remove 1000 rows with missing InternetService values from the working cleaned dataset. The original dataset was 7,043 rows, and removing those rows resulted in 6,042 rows in the cleaned version. This is not really a limitation, but removing the data helped clean the data correctly because it was hard to assume what to fill the data with, and it would be best to delete it.

I worked on converting tenure, Monthly Charges, and Total Charges into numeric values. Blank Total Charges entries become missing values during numeric conversion. I used medians to fill out missing numerical values and mode to fill the selected missing categorical values. The final workflow should be able to calculate these values from the training data only.

I worked on encoding the categorical data for machine learning. This included mapping selected Yes/NO to 1 and 0, converting Churn to numerical with No to 0 and Yes to 1, encoding Contract to represent month-to-month, one-year, and two-year contracts, and using one-hot encoding for categorical fields such as Payment Method and Internet Service. I also revised the Payment Method encoding to preserve all four payment-method categories.

Toan- Training/Test Workflow and Class Balancing

One of my contributions to this project was preparing the one-page PDF document describing our dataset and what we wanted to accomplish with it. I worked with the Telco Customer Churn dataset, which contains customer information such as tenure, monthly charges, contract type, and services. Our goal was to prepare this data for a machine learning project that could predict whether a customer would leave the company. Working on the proposal helped me understand why choosing the right dataset and defining a clear project goal are important before building a model.

I also worked on organizing our Jupyter notebook and making the workflow easier to follow. At first, I had trouble importing the dataset from Kaggle into Google Colab. After trying different methods, I decided to upload it to Google Drive and set up the notebook to load it directly from there. This made it easier for our team to access the dataset without uploading it again every time we started a new session. While reviewing the notebook, I also noticed duplicate records, missing values, and blank entries that needed to be checked. I helped reorganize the code and create visualizations so we could better understand the data and see how it changed during preprocessing.

Another thing I learned was the importance of separating training data from testing data. Our notebook uses a 70/30 split with stratification to keep the proportions of customers who stayed and left relatively similar in both sets. I also reviewed the class-balancing process using RandomOverSampler and the importance of checking the number of churn and non-churn cases before and after balancing. Oversampling should only be applied to the training data, while the testing data should keep its original class distribution.

While comparing our notebook with the instructor's requirements, I noticed that some EDA steps were performed before the data split. This showed me that we still needed to review the workflow, confirm the EDA requirements with the instructor, and check for possible data leakage. Overall, this project helped me understand how dataset selection, data cleaning, visualization, and a proper training/testing workflow work together to prepare data for a reliable machine learning model.

Montserrat: Visualization, Output Files, and Submission Organization

I worked on preparing and inspecting the Telco Customer Churn dataset using Python in Google Collab. My contribution focused on organizing the notebook, understanding the dataset, checking data quality, and documenting the code. This initial work helped establish a foundation for the later stages of data preprocessing and machine learning.

Throughout this process, I added explanatory comments to the data. These comments describe the purpose of the code, explain important operations, and make the workflow easier to

follow. I also reviewed the missing-value comparison code to understand how to evaluate data quality before and after imputation.

One challenge I faced was understanding the purpose of each code section and how the different steps connect. I learned that loading a dataset is only the beginning of a machine learning project. The data must also be inspected, checked for missing values, and prepared carefully before building machine learning models. I also learned why preserving an original copy of the dataset is important and why the data should be divided into training and testing sets for model development and evaluation.

Overall, I earned the importance of writing clear comments and organizing code so that the analysis is easier to understand and reproduce. These skills will help me as I continue working on data preprocessing and machine learning models in the project.

4. Shared Team Responsibilities

- Review the final notebook and confirm that all cleaning choices match the actual code and the instructor's requirements.
- Make all data changes through Python in Jupyter rather than manually editing the dataset.
- Confirm that the split is 70% training and 30% testing.
- Ensure EDA is performed only on the training data and clarify the instructor's requirement that the test dataset remains untouched.
- Add the required Matplotlib distribution plots and before-and-after sample outputs.
- Run the notebook from a fresh runtime, fix errors, and generate the required output files through Python.

- Complete each member’s own reflection and accurately document each member’s contributions.
- Confirm the dataset approval and exact Kaggle URL.
- Prepare the one-page proposal PDF.
- Upload the notebook, journals, dataset URL document, and final cleaned dataset to GitHub.
- Submit the GitHub repository URL in Canvas.